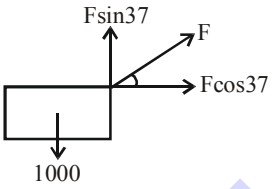
SOLUTION
SECTION-I

Single Correct Answer Type **5 Q. [3 M (–1)]**

1. Ans. (B)

Sol. $F\cos 37 = \mu\left(1000 - \frac{3F}{5}\right)$



$\frac{4}{5}F = \frac{1000}{3} - \frac{F}{5}$

$F = \frac{1000}{3}$

$\frac{1000}{3} = 25(g + a)$

$a = \frac{10}{3}$

2. Ans. (B)

Sol. Tension = T – mg = ma

T = m(g + a)

T = 10(10 + 2)

T = 10 × 12 = 120

Upper support 120 + 80 = 200

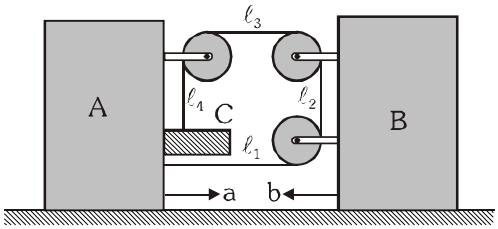
(80) weight of latter monkey

3. Ans. (A)

Sol. Acceleration of C in horizontal direction = acceleration of A=a (right direction) = $a\hat{i}$

Accelration of C in vertical downward direction = $2(a+b)\hat{j}$

Hence, net acceleration of C = $a\hat{i} - 2(a+b)\hat{j}$ (see figure)



From length constraint $\ell_1 + \ell_2 + \ell_3 + \ell_4$=C

$\ddot{\ell}_1 + \ddot{\ell}_2 + \ddot{\ell}_3 + \ddot{\ell}_4 = 0$ –a –b + 0 –a –b + c = 0

c = 2a + 2b

From wedge constraints, acceleration of C towards right side is a. Acceleration of C w.r.t. ground

$= a\hat{i} - 2(a+b)\hat{j}$

4. Ans. (A)

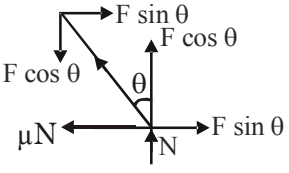
Sol. N = F cos θ

F sin θ = μN

$\frac{\sin\theta}{\cos\theta} = \mu$

tan θ = 0.9

θ = $\tan^{-1}$ (0.9)

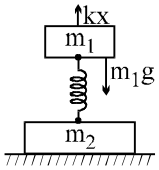


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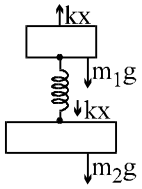
5. Ans. (A)

Sol. Initially $m_1 g = kx$

When support is removed, spring force does not change.



New FBD



For m_1 : $m_1 g - kx = m_1 a_1$ $\Rightarrow a_1 = 0$

For m_2 : $m_2 g + kx = m_2 a_2$ $\Rightarrow a_2 = \dfrac{(m_1 + m_2)g}{m_2}$ ↓

Multiple Correct Answer Type 5 Q. [4 M (–1)]

6. Ans. (A,B)

Sol. Man $N + T = mg$

$N + T = 500$ .....(i)

Weghing machine $T = mg + N$

$T - N = mg$

$T - N = 200$ ....(ii)

$N + T = 500$

$T - N = 200$

$2T = 700$

$T = 350$

$N + 350 = 500$

$N = 150$

Reading 15 kg

$T = 350$



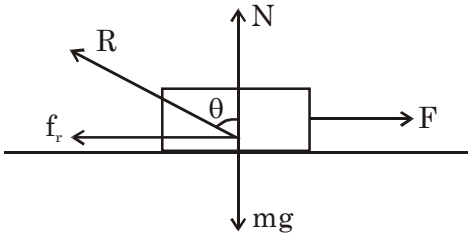
7. Ans. (A,B,C)

Sol. Net force by surface is resultant (R) of normal and friction

$\tan\theta = \dfrac{f_r}{N} = \dfrac{f_r}{mg}$

for $F \geq \mu mg$ $f_r = \mu mg = \dfrac{1}{\sqrt{3}} mg$

$\tan\theta = \dfrac{\frac{1}{\sqrt{3}} mg}{mg} = \dfrac{1}{\sqrt{3}} \Rightarrow \theta = 30°$



For $F < \mu mg$ $f_r = F$ and $\tan\theta = \dfrac{F}{mg} < \dfrac{\mu mg}{mg}$

$\tan\theta < \dfrac{1}{\sqrt{3}} \Rightarrow \theta < 30°$

(D) For $F < \mu mg$, θ depends on m but for $F \geq \mu mg$ value of θ is fixed ($= 30°$) and does not depend on m

8. Ans. (A,B,C)

Sol. $F - f = ma$

$30 - 10 = m \times 10$ $\mu mg = 10$

$20 = 10m$ $\mu 20 = 10$

$m = 2kg$ $\mu = \dfrac{1}{2}$

block start at force = 10. Limiting friction is 10.

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9. Ans. (B,C,D)

Sol. Spring force does not change instantaneously.

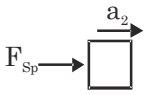
Thus for m_1 ; $a_1 = a_0$

for m_2 $F_{S_p} = m_2 a_2$ .....(i)

instantaneously aftr F_2 is withdrawn

Initially $F_{S_p} - F_2 = m_2 a_0$

$F_{S_p} = F_2 + m_2 a_0$ .....(ii)



from (i) and (ii) $a_2 = \frac{F_2}{m_2} + a_0$

10. Ans. (A,B,C)

Sol. Mass 2 kg is at rest, so $T_2 = 20N$

So for (2)

$T_2 - mg = ma$

$20 - 10 = 1 \times a$

$a = 10\ m/s^2$

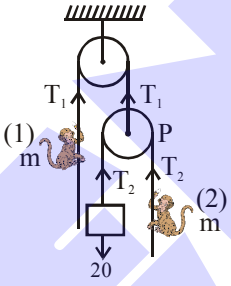
$T_1 = 2\ T_2 = 40\ N$

So for (1)

$T_2 - mg = ma$

$40 - 10 = 1 \times a$

$a = 30\ m/s^2$



End of rope is coming down with $5\ m/s^2$

So acceleration of (1) w.r.t. his rope is

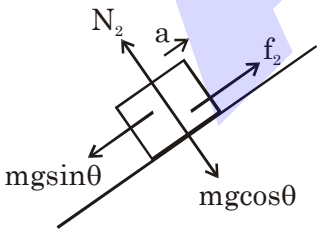
$30 + 5 = 35\ m/s^2$ upwards

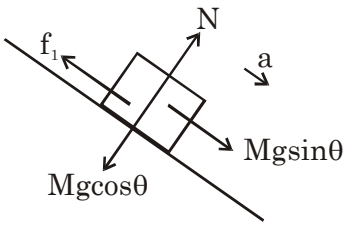
Acceleration of monkey (2) with respect to his rope is zero because he is just holding the string.

Linked Comprehension Type (2 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

11. Ans. (D)

Sol. No slipping between ribbon and blocks implies ribbon also moving with blocks and blocks and ribbon have acceleration of same magnitude





As ribbon is massless, net force on it is zero

$\Rightarrow f_1 = f_2 = f$

by F = ma $Mg\sin\theta - f = Ma$

$f - mg\sin\theta = ma$

$$\Rightarrow a = \frac{g\sin\theta(M-m)}{M+m}$$

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12. Ans. (C)

Sol. Tension = f

$$Mg\sin\theta - f = Ma = \frac{Mg\sin\theta(M-m)}{M+m}$$

$$f = \frac{2Mmg\sin\theta}{(M+m)}$$

13. Ans. (A)

Sol. As θ increases $\Rightarrow$ f increases

as value $f_1)_{max} > f_2)_{max}$

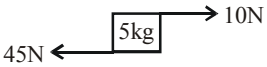
So first m block slip on ribbon

14. Ans. (A)

Sol. At F = 10N, motion between blocks starts

15. Ans. (C)

Sol.



So the block will never move

16. Ans. (A)

Sol. 30 – 10 = 10 a

20 = 10 a

a = 2

Matching List Type (4 × 4) 1 Q. [3 M (–1)]

17. Ans. (D)

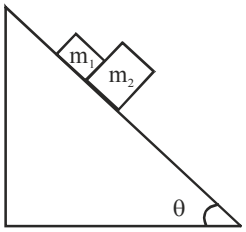
Sol. The system slip down if

$(m_1 + m_2)\, g \sin\theta > \mu m_2 g\cos\theta$

$$\tan\theta > \frac{\mu m_2}{m_1+m_2} > \frac{0.3\times 2}{3}$$

$\tan\theta > 0.2$

$\Rightarrow \theta > 11.5°$



For P and Q system will remain stationary hence friction = $(m_1 + m_2)\, g \sin\theta$

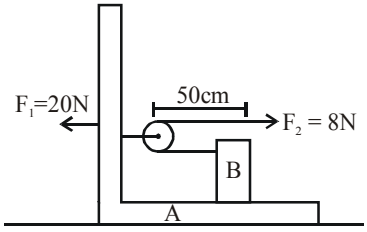
For R and S system will move hence limiting friction acts friction = $\mu m_2 g \cos\theta$

SECTION-II

Numerical Answer Type Question (upto second decimal place) 2 Q. [3(0)]

18. Ans. 0.5 s

Sol.



$$a_A = \frac{20-16}{1} = 4 m/s^2$$

$$a_B = \frac{8}{1} = 8 m/s^2$$

$a_{B/A} = 4 m/s^2$

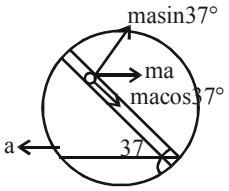
$0.5 = 1/2 \times 4t^2$

$$t = \frac{1}{2}\text{sec} = 0.5\,\text{sec}$$

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19. Ans. $10\ m/s^2$

Sol.



$ma\cos 37 - \mu ma\sin 37 - \mu mg = ma_0$

$$25\times\frac{4}{5}-\frac{2}{5}\times 25\times\frac{3}{5}-\frac{2}{5}\times 10 = a_0$$

$20 - 6 - 4 = 90$

$a_0 = 10\ m/s^2$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **4 Q. [4 M (0)]**

20. **Ans. 8**

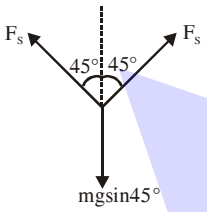
Sol. $a = -3$

$$u=\sqrt{\left(25\times\frac{4}{5}-8\right)^2+\left(25\times\frac{3}{5}\right)^2}=\sqrt{(12)^2+(15)^2}=3\sqrt{41}$$

at $t=\frac{3\sqrt{41}}{3}$ so it will stop relative motion before 10 sec. So it will have same velocity as plank.

21. **Ans. 5**

Sol.



$$\frac{2F_s}{\sqrt{2}}=\frac{mg}{\sqrt{2}};\ F_s=\frac{mg}{2}\quad\therefore \text{ acceleration}=\frac{F_s}{m}=\frac{g}{2}=5\ m/s^2$$

22. **Ans.** **5**

Sol. The girl will catch the ball when their horizontal displacement becomes equal

$$\therefore 12t=\frac{1}{2}\left(g\sin 37^\circ\cos 37^\circ\right)t^2\Rightarrow t=5\sec$$

23. **Ans. 0**

Sol. $f_k = 6$ N which will act till velocity is not zero.



Let velocity becomes zero at t, then $0 - 4.5 = \int_0^t (3-3t)dt \Rightarrow t = 3$ so at t = 3, v = 0

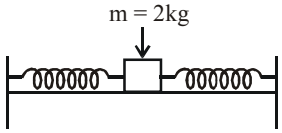
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SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

24. Ans. (A) - (S) ; (B) - (S) ; (C) - (Q) ; (D) - (T)

Sol.



$\text{force} = 100\times\frac{2}{100}+100\times\frac{2}{100}$

$4 = 2a$; $a = 2$

(C) $100\times\frac{4\times1}{100}+100\times\frac{4}{100}$

$8 = 2a$; $a = 4$

(D) $100\times\frac{1}{100}+100\times\frac{1}{100}$

$2 = 2a$; $a = 1$

Subjective Type **6 Q. [4 M (0)]**

25. Ans. 3/4

Sol. If m notsliding

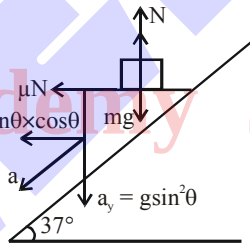
$a = g\sin\theta$

$mg - N = mg\sin^2\theta$

$N = mg(1-\sin^2\theta) = mg\cos2\theta$

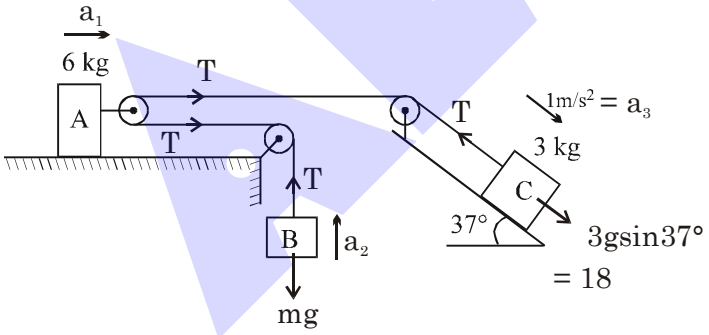
$\mu N = mg\sin\theta\cos\theta$

$\mu = \frac{\sin\theta}{\cos2\theta} = \tan\theta = \frac{3}{4}$



26. Ans. $M_B = 15$ kg, $T = 15$, $a_A = 5\ m/s^2$, $a_B = 9\ m/s^2$

Sol.



By string constrain

$2Ta_1 + Ta_2 - Ta_3 = 0$

$2a_1 + a_2 = 1$ .... (i)

by F = ma

$18 - T = 3\times1 \Rightarrow T = 15$

$2T = 6a_1 \Rightarrow a_1 = \frac{2\times15}{6} = 5m/s^2$

by equation (i) $2(5) + a_2 = 1 \Rightarrow a_2 = -9$

$T - mg = ma_2 \Rightarrow 15 - 10\,m = m(-9)$

$\Rightarrow m = 15$ kg

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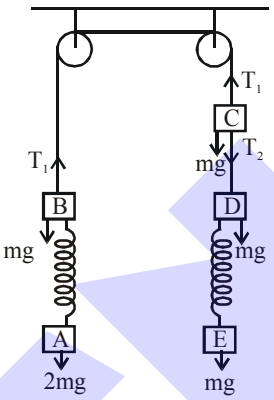
27. **Ans.** (i) $a_A = g\downarrow,\ a_B = \frac{2g}{3}\uparrow,\ a_c = \frac{2g}{3}\downarrow,\ a_D = \frac{2g}{3}\downarrow,\ a_E = 0$

(ii) $a_A = 0,\ a_B = \frac{g}{3}\downarrow,\ a_C = \frac{g}{3}\uparrow,\ a_D = \frac{g}{3}\uparrow,\ a_E = g\downarrow$

(iii) $a_A = 0, a_B = g\downarrow, a_C = g\uparrow, a_D = 2g\downarrow, a_E = 0$

(iv) $a_A = 0, a_B = 3g\downarrow, a_C = \frac{3g}{2}\downarrow, a_D = \frac{3g}{2}\downarrow, a_E = 0$

Sol. (i) spring 1 is cut



$\vec{a}_A = g\downarrow$

acceleration of B,C,D are a.

$mg - T_1 = ma$

$T_1 - mg - T_2 = ma$

$T_2 - 2mg = ma$

$-2mg = 3ma$

$a = \frac{2g}{3}$

$a_B = \frac{2g}{3}\uparrow$ $a_C = \frac{2g}{3}\downarrow$ $a_D = \frac{2g}{3}$

$a_E = 0$

(ii) spring (2) is cut $a_A = 0$

all of B.C.D. are let a

$3mg - T_1 = ma$ $\vec{a}_B = \frac{g}{3}\downarrow$ $\vec{a}_C = \frac{g}{3}\uparrow$

$T_1 - T_2 - mg = ma$ $\vec{a}_D = g/3\uparrow$

$T_2 - mg = ma$ $\vec{a}_E = g\downarrow$

$a = \frac{g}{3}$

(iii) String between C & D is cut

$a_A = 0$ $a_E = 0$ $a_D = 2g\downarrow$

let acceleration of B & C

$3mg - T_1 = mg$ $a_B = g\downarrow$ $a_C = g\uparrow$

$T_1 - mg = ma$

(iv) Between B & C is cut

$a_A = 0$ $a_E = 0$

For a_B

$3mg = ma_B$

$a_B = 3g\downarrow$

acceleration of C & D are (a) (let)

$2mg - T = ma$

$mg + T = ma$

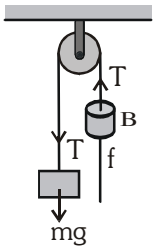
$a = \frac{3g}{2}$ $a_c = \frac{3g}{2}\downarrow$ $a_D = \frac{3g}{2}\downarrow$

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28. **Ans.** 5 m/s^2 downwards, 0 m/s^2, 10 N

Sol. $20 - T = 2a$



for sleeve

$10 - f = 1a_1$

$T = f$

$a_1 + a = 5$

$a_1 = 5 - a$

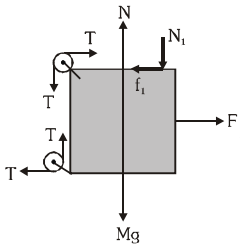
$20 - f = 2a$ acceleration objective

$10 - f = 5 - a$ $a_1 = 5 - 5 = 0$

$10 = 3a - 5$ $20 - T = 2 \times 5$

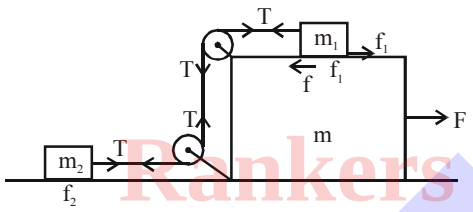
$a = 5$ m/s^2 downwards $T = 10$N

29. **Ans.** (i)



(ii) a = 3/5 m/s^2 , T = 18 N, F = 60N

Sol.



$f_1 = 2f_2$

for bigger block

$f_1 - T = 20a$

$F - f_1 = 50\,a$ $T - f_2 = 5a$

$F - 2f_2 = 50\,a$ ....(i)

$2f_2 - T = 20a$ ...(ii)

$T - f_2 = 5a$ ....(iii)

$f_2 = 25\,a \Rightarrow 15 = 25\,a \Rightarrow a = a = \frac{3}{5} m/s$

from equation (i) $F_2 - 50a = 50\,a$

$F_2 = 100\,a = 60$ N

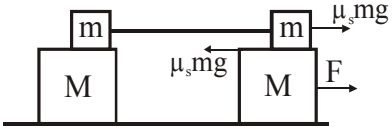
$f_2 = \mu\, m_2 g = 0.3 \times 50$ $T - 15 = 3$

$= 15$ N $T = 18$ N

30. **Ans.** $\mathbf{F = 2\mu_s mg}\left(\frac{m+M}{2m+M}\right)$

Sol. $\mu_s mg = (M+2m)a \Rightarrow a = \frac{\mu_s mg}{M+2m}$

Also $F - \mu_s mg = Ma \Rightarrow 2\mu_s mg \frac{(m+M)}{(M+2m)}$



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